PawWatch

Edge-AI Cat Health Monitor

Link to GitHub Repository

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Problem

Cat hides illness

Cats instinctively hide pain and illness; obvious symptoms such as vomiting may appear only when it is more severe [1], [2]. Earlier signs show up in daily habits: vets list changes in appetite, thirst and litter box use as warning signs [2], and each common disease changes them in its own way:

Disease	How common	Eating	Drinking	Litter
Kidney disease	Up to 30–40% of older cats may have it; only 3.6% are diagnosed [3], [4]		↑	↑
Hyperthyroidism	8.7% of cats over 10 [5], [6]	↑	↑	↑
Diabetes	About 1 in 200 cats [7]	↑	↑	↑
Urinary tract disease	A leading reason cats visit vets [8]			↑

A 2026 study of 136 cats found those with kidney disease used the litter box about twice as often, even in early stages [9]. A blocked urinary tract may kill within 24–48 hours; its first sign is many litter box trips [8].

Solution

Today’s smart devices each watch one spot

Each smart device on the market watches one spot, so watching food, water and litter takes several devices and apps.

	Food	Water	Litter
Smart feeders	✓	✗	✗
Smart water fountains	✗	✓	✗
Smart litter boxes	✗	✗	✓
Pet cameras (video only, no visit log)	✗	✗	✗
PawWatch	✓	✓	✓

However, illness shows up as a **combination**. For example, kidney disease raises both drinking and litter box use [3], so a water-only or litter-only device sees half the pattern. Once set up, PawWatch watches all three spots with one box, and the video never leaves home.

Solution: PawWatch

Owners are out all day, and no single device sees the whole pattern. **PawWatch** is a vision-based monitoring system: low-cost cameras watch the food bowl, water bowl and litter box, and an on-device AI accelerator recognizes the cat and logs every visit. Video never leaves home.



What it records

- **Visit log:** how often and how long the cat eats, drinks and uses the litter box
- **No-go zone alarm:** plays a sound when the cat jumps onto the counter

What it tells you

- **Anomaly alert:** ≥ 2 or $\leq \frac{1}{2}$ times of the cat's 7-day average
- **Daily report:** a plain-language summary
- **Dashboard:** today's counts, timeline and 7-day trends

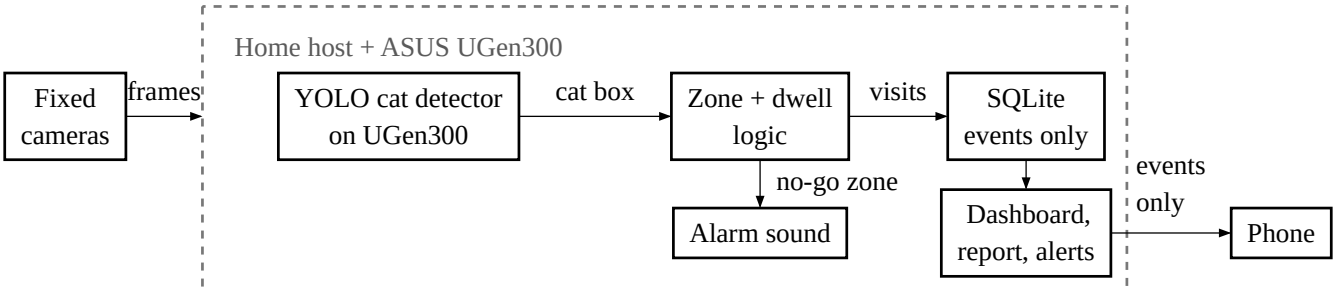
Why run AI at home, not in the cloud

	Cloud pet camera	PawWatch
Who sees your home video?	It's uploaded to the company's servers. At one home-camera maker, staff watched customers' private videos [10]	No one. Video never leaves home; only events like "14:32, drank for 25 s" are stored
Monthly fee	Smart features usually need a paid subscription	None. No servers to pay for
Internet goes down	Cloud features stop until it's back	Keeps logging; alerts are sent once it's back
Cat jumps on the counter?	Frames go to the server and back first	The alarm sounds right away, from the box at home

Running 24/7 is cheap: the AI accelerator draws 2.5 W, less than an LED bulb [11].

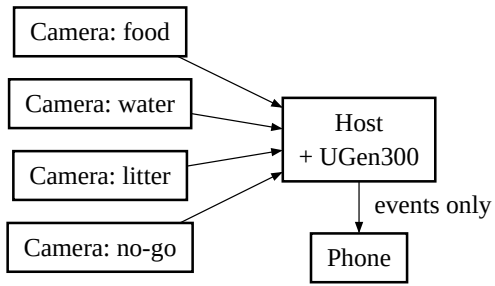
PawWatch System

Architecture: video in, events out



- **Detect:** the host decodes each camera's video and the UGen300 runs a stock YOLO model that finds the cat. Frames live only in memory and are dropped after inference; nothing saves or uploads them.
- **Turn boxes into visits:** the cat's box is matched to the zones drawn for that camera (food, water, litter, no-go), and a visit is logged once the cat stays long enough. A no-go zone rings the alarm right away.
- **Store and report:** each visit is one row in a local SQLite database. The dashboard, anomaly alert and daily report read only these rows, so the only thing that can leave the home is an event like "07:30, ate for 31 s".

Hardware: one host, many fixed cameras



We mount **fixed cameras**, one at each spot we watch: the food bowl, the water bowl, the litter box and any no-go spot such as the kitchen counter. Each camera streams its video over the home network to **one small host computer** (a mini PC or Raspberry Pi).

The **ASUS UGen300** is plugged into the host over USB or M.2. The host decodes the video and hands frames to the UGen300, whose Hailo-10H chip (40 TOPS at 2.5 W [11]) runs the cat detector. The host then turns detections into visits, stores them and serves the dashboard. Only these events reach the owner's phone.

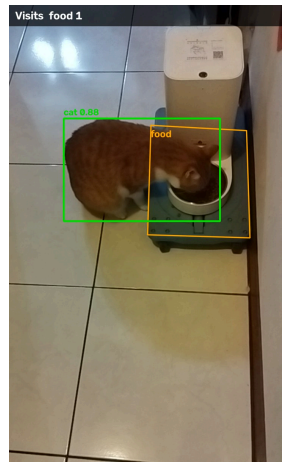
Why fixed cameras: zones are drawn in pixels, so a camera that never moves needs them drawn only once, at setup. One camera per spot also keeps the cat large in the frame, which makes detection reliable.

Detection: from pixels to “the cat is at the bowl”

A stock YOLO model, no training. We use YOLOv8n pretrained on the public COCO dataset, which already has a “cat” class. We keep only that class (score ≥ 0.4). No cat photos were collected and no model was trained or fine-tuned, so any home can use it out of the box.

Zones are drawn once. At setup, the owner clicks a polygon around each spot on a still frame (`draw_zones.py`); it is saved as one small JSON file per camera.

Which zone is the cat in? We take the highest-scoring cat box and measure how much of it overlaps each zone, as a share of the **smaller** of box and zone. The zone with the most overlap wins, if it reaches 20%. Measuring against the smaller shape lets a zone hug a small bowl, and a box that only grazes a zone’s edge doesn’t count.



Detection window: cat box, food zone and visit count

Event logic: from detections to visits

The event logic is pure code: it takes $(\text{video time}, \text{zone})$ pairs and returns visits, so it is tested without a model or video. Timestamps are **video time** (recording start + frame offset), so recorded footage replays exactly.



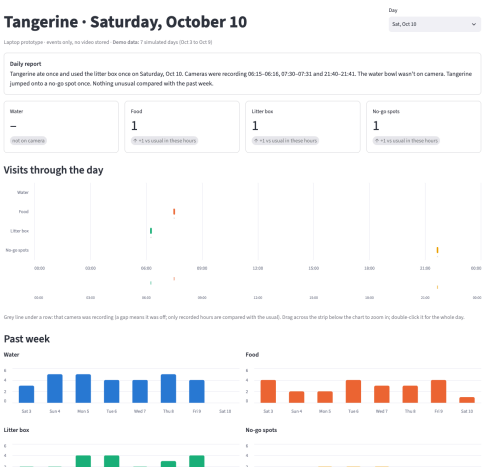
Minimum dwell	A visit counts after 5 s in one zone, so walking past a bowl isn’t a meal. No-go zones need only 1 s: a quick jump still counts.
Gap tolerance	The cat may vanish for up to 5 s (turning, hidden by the litter box rim) without splitting one visit into two.
Live writes	A visit is stored as soon as it qualifies and its end moves forward while the cat stays, so the dashboard follows a live camera.
Alarm	Every frame in a no-go zone calls the alarm at once; a 30 s cooldown gives one sound per jump, not per frame.

Storage and outputs: what the owner sees

One table is the whole record. Each visit is one row in local SQLite: `camera, zone, start, end, duration, simulated`. A second table logs when each camera was recording, so hours with the camera off are never read as “didn’t drink”.

Anomaly alert. For each of food, water and litter, today’s count is compared with the cat’s own average over the previous 7 days, **over the same hours**, so a normal morning isn’t flagged for a missing dinner. An alert needs both $\geq 2\times$ or $\leq ½$ the average **and** a difference of at least 3 visits, so 1 vs. 0 is not an alert.

Daily report and alarm. Template sentences turn the counts into plain language. Wording suggests watching the cat or seeing a vet, never a diagnosis. No-go jumps play a sound on the host and can send a text message (never a frame).



Dashboard: today’s counts, timeline, 7-day trend

Impact and Business

Expected benefits

TODO: 目前是 AI 生的

- **Owners:** catch changes early instead of when symptoms are obvious
- **Vets:** objective at-home records after surgery or before a follow-up

Without PawWatch

The cat uses the litter box twice as often while the owner is at work. Nobody notices until symptoms are obvious.

With PawWatch

Litter box visits reach 2× the 7-day average: an alert the same day suggests watching the cat or seeing a vet, with the daily report in hand.

Market: more cats in Taiwan, in a growing pet tech market

- **More cats:** Taiwan now has 1.74 M pet cats, up 33% in two years, and more cats than dogs [12]
- **Growing pet tech:** the global market may grow from US$15.6 B (2025) to US$52.9 B (2035) [13]

Business model: ride on hardware people already buy

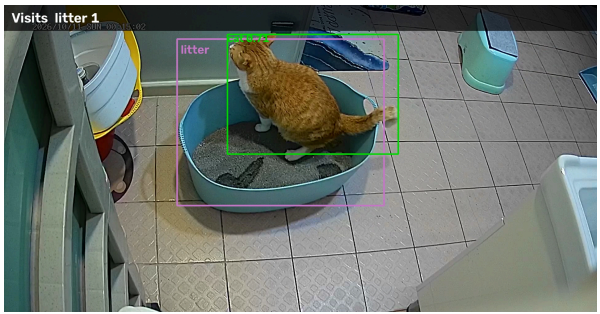
TODO: 目前是 AI 生的, 完全不知道這啥

Customer	Offer	Revenue
Device makers (pet cameras, home hubs, PCs with edge AI)	PawWatch edge software on their hardware: no cloud cost for them	Per-unit license
Vet clinics	At-home monitoring kits after surgery and for chronic cases (e.g. kidney disease)	Monthly rental per kit
Cat owners	Box + cameras, no subscription	One-time purchase

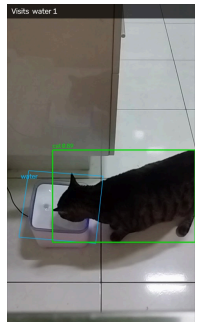
- Licensing spreads the UGen300 cost: it doesn’t land on one product
- One UGen300 serves every camera in the home, so each extra spot is only a camera

Appendix

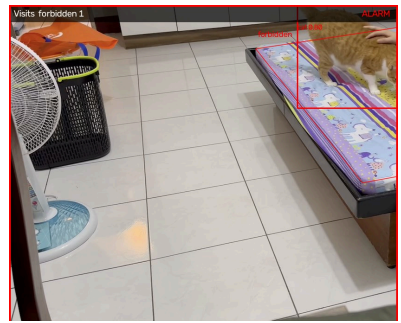
Detection screenshots



Tangerine in the litter zone



Ami at the water bowl



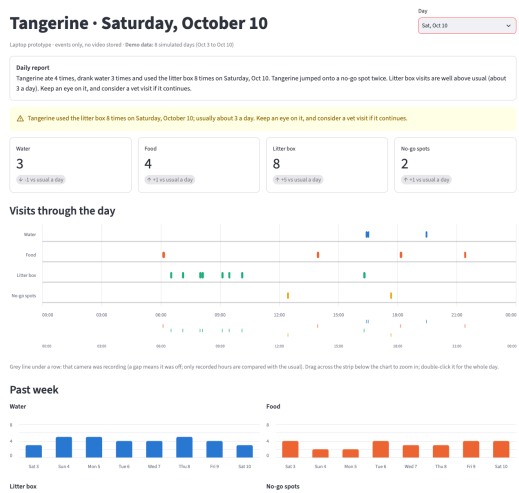
No-go zone: box turns red, alarm rings

Each window shows the cat box with its score, the camera’s zones and the visits counted so far. The windows are only displayed; frames are never written to disk.

Dashboard with an anomaly alert (simulated data)

This dashboard runs on **simulated** data only, to show what an alert looks like: a day where the litter box count reaches at least twice the cat's 7-day average.

The alert names the change and the usual count, and suggests keeping an eye on the cat or seeing a vet. It never names a disease.



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